EMILIE BURNHAM

PhD Student in Astronomy and Astrophysics
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RESEARCH INTERESTS

Galaxy formation and evolution; large-scale surveys; scientific software development; machine-learning; stellar population synthesis; SED fitting; cosmological simulations.

EDUCATION

The Pennsylvania State University **2023–2029**
PhD, MS in Astronomy & Astrophysics State College, PA, US

- Thesis Advisor: Dr. Joel Leja.
- Expected graduation date: May 2029.

Texas Christian University **2019–2022**
BSc in Physics & Astronomy Fort Worth, TX, US

- Thesis Advisor: Dr. Mia Bovill.
- Graduated summa cum laude.

RESEARCH EXPERIENCE

Penn State University **2023–**
Graduate Research Assistant State College, PA, US

Developed tools (ESPS, PIGLET) to enable fast modeling of galaxy spectra in large surveys. Applied these tools to infer the recent star formation histories of galaxy populations across cosmic time, with a particular emphasis on burstiness. Presented results at conferences and seminars, and published research in peer-reviewed journals.

Cosmic Dawn Center (DAWN), Neils Bohr Institute, University of Copenhagen **2026**
Copenhagen, Denmark
NSF DAWN IRES Senior Scholar

Mentored undergraduate students participating in the program and led workshops on research skills used in the field of galaxy formation and evolution. Continued thesis research while in residence at DAWN.

University of Colorado at Boulder **2021**
NSF REU Research Assistant Remote

Conducted research on how well modern radiative transfer codes can recover the absorption features in different spatial features (e.g., plage, faculae) of the solar disk. Presented results at the 2021 Winter American Geophysical Union meeting.

Texas Christian University **2020–2023**
Undergraduate Research Assistant Fort Worth, TX, US

Conducted research on modeling the observable affects of a warm dark matter cosmology in the JWST era. Reduced spectra of nearby open clusters to calculate alpha element abundances. Presented results at the 2023 Winter American Astronomical Society meeting.

INDUSTRY EXPERIENCE

Front Line Mobile Health, PLLC **2023–2024**
Clinical Data Analyst Georgetown, TX, US

Conducted research on the health of first responders. Developed a data pipeline to analyze and visualize health data to employers of first responders.

TEACHING EXPERIENCE

Penn State University **2026**
Undergraduate Research Mentor, Students Together for Astronomical Research (STAR) Program State College, PA

Penn State University **2025**
Guest Lecturer, ASTRO 291: Astronomical Methods and the Solar System State College, PA

Texas Christian University **2023**
Teaching Assistant, PHYS 10293: Archeoastronomy Fort Worth, TX

Texas Christian University **2022**
Teaching Assistant, PHYS 20475: Physics I for Majors Fort Worth, TX

FIRST-AUTHOR PUBLICATIONS

[1] Emilie Burnham, Bingjie Wang, Joel Leja, Owen Gonzales, Jenny E Greene, Kartheik G Iyer, Abby Mintz, David J Setton, Sarah Wellons, Rachel Bezanson, and others. *It's More Complicated Than You Think: A Forward Model to Infer the Recent Star Formation History, Bursty or Not, of Galaxy Populations* The Astrophysical Journal (2026).

CO-AUTHOR PUBLICATIONS

[1] Cleri, Nikko J, Olivier, Grace M, Backhaus, Bren E, Leja, Joel, Papovich, Casey, Trump, Jonathan R, Arrabal Haro, Pablo, Buat, Veronique, Burgarella, Denis, Burnham, Emilie, others. *Optical Strong Line Ratios Cannot Distinguish between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities* The Astrophysical Journal (2025).

[2] Wang, Bingjie, Leja, Joel, Atek, Hakim, Bezanson, Rachel, Burnham, Emilie, Dayal, Pratika, Feldmann, Robert, Greene, Jenny E, Johnson, Benjamin D, Labbe, Ivo, others. *Population Models for Star Formation Timescales in Early Galaxies: The First Step toward Solving Outshining in Star Formation History Inference* The Astrophysical Journal (2025).

[3] Levitt, Danielle E, Wohlgemuth, Kealey J, Burnham, Emilie, Conner, Michael J, Collier, J Jason, Mota, Jacob A. *Hazardous alcohol use and cardiometabolic risk among firefighters* Alcohol: Clinical and Experimental Research (2025).

[4] Levitt, Danielle E, Burnham, Emilie, Conner, Michael J. *At-risk alcohol use and cardiometabolic risk factors among first responders in the US* Drug and Alcohol Dependence (2024).

[5] Mintz, Abby, Setton, David J, Greene, Jenny E, Leja, Joel, Wang, Bingjie, Burnham, Emilie, Suess, Katherine A, Atek, Hakim, Bezanson, Rachel, Brammer, Gabriel, others. *Taking a Break at Cosmic Noon: Continuum-selected Low-mass Galaxies Require Long Burst Cycles* The Astrophysical Journal (2026).

[6] Wohlgemuth, Kealey J, Conner, Michael J, Burnham, Emilie, Mota, Jacob A. *Occupational Health Disparities: A Profile of Firefighters and Police Officers* The Journal of Strength and Conditioning Research (2026).

[7] Mitsuhashi, Ikki, Suess, Katherine A, Leja, Joel, Bezanson, Rachel, Greene, Jenny E, Burnham, Emilie, Khullar, Gourav, Mintz, Abby, Nanayakkara, Themiya, Glazebrook, Karl, others. *UNCOVER/MegaScience Finds Uniform and Highly Bursty Star Formation at $3 < z < 9$, consistent with the High-Redshift UV Luminosity Function* arXiv preprint arXiv:2601.16284 (2026).

[8] Cleri, Nikko J, Lewis, Zach J, Leja, Joel, Helton, Jakob M, Burnham, Emilie, Curtis, Olivia, de Graaff, Anna, Hirschmann, Michaela, Katz, Harley, Maseda, Michael V, others. *RUBIES: The Evolution of the Ionization Parameter from 0< z< 9* arXiv preprint arXiv:2605.30410 (2026).

HONORS AND AWARDS

Harold and Nancy O'Connor Graduate Excellence Fund Scholarship **2025**
Penn State Eberly College of Science

Frymoyer Award **2025**
Institute of Gravitation and the Cosmos

Zaccheus Daniel Foundation for Astronomical Science Scholarship **2025**
Penn State Department of Astronomy and Astrophysics

University Graduate Fellowship **2024**
Penn State University

Stephen B. Brumbach Distinguished Graduate Fellowship in Science **2023**
Penn State University

Verne M. Willaman Distinguished Graduate Fellowship in Science **2023**
Penn State University

Homer F. Braddock Scholarship in Biology, Chemistry, and Physics **2023**
Penn State University

Dr. C.A. Quarles Physics and Astronomy Scholarship **2021**
Texas Christian University Department of Physics and Astronomy

Joseph Morgan Physics Scholarship **2021**
Texas Christian University Department of Physics and Astronomy

Bob Bolen Leadership Scholarship **2021**
The Sarah & Ross Perot Jr. Foundation, Texas Christian University

TCU Scholar **2021**
Texas Christian University

College of Science and Engineering Dean's List **2019**
Texas Christian University College of Science and Engineering. Six-time awardee.

LEADERSHIP AND SERVICE

Astronomy on Tap State College Organizer **2024–**
Astronomy on Tap State College State College, PA
Organized monthly public talks on astronomy topics, coordinated with speakers, and promoted events to engage the local community in astronomy outreach.

Society of Physics Students (SPS) President and Treasurer **2021–2022**
Texas Christian University Chapter of the Society of Physics Students Fort Worth, TX
Led the student chapter of the Society of Physics Students, organized events and meetings, managed finances, and coordinated events to grow participation across departmennts.

OUTREACH

Public Lecture on "Why is Astronomy a Big Data Field?" **2025**
Astronomy on Tap State College State College, PA

Public Lecture on "The Milky Way Through the Rainbow" **2024**
Astro Night at Penn State State College, PA

Public Lecture on "Suprises at the Edge fo the Universe Found with JWST" **2024**
Central Pennsylvania Observer's Society State College, PA

PROFESSIONAL PRESENTATIONS

Invited Talks

- *"It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"*
  - Penn State University Center for Astrostatistics and Astroinformatics Lunch Seminar | State College, PA | 2026
  - University of Copenhagen Cosmic Dawn Center (DAWN) Cake Talk Seminar | Copenhagen, Denmark | 2026
  - University of Pittsburgh Joint Extragalactic Group Meeting | Pittsburgh, PA | 2025
  - Penn State University Institute for Gravitation and the Cosmos Primordial Universe and Gravitation Seminar | State College, PA | 2025
  - Princeton University Galactic/Extragalactic Reading Group | Princeton, NJ | 2025

Contributed Talks

- *"It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"*
  - Eberly Research Showcase | State College, PA | 2025 | (First-place prize)
  - Penn State University Department of Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2025
  - 7th Neighborhood Workshop on Astrophysics and Cosmology | State College, PA | 2025 | (First-place prize)
- *"The Need for Speed in Galaxy Evolution Surveys"*
  - Penn State Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2025
- *"Measuring the Family Histories of Bursty Star Formation within Galaxy Populations"*
  - Penn State Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2024
- *"Forensic Astronomy"*
  - Texas Christian University Annual Student Research Symposium | Fort Worth, TX | 2021

Contributed Posters

- *"It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"*
  - Carnegie Mellon University Statistical Methods for the Physical Sciences Research Center Workshop | Pittsburgh, PA | 2025 | (simulation-based inference)
  - Penn State University Institute for Computational Data Sciences Symposium | State College, PA | 2025 | (computational methods)
- *"Warm or Cold Dark Matter? A Love-Heat Relationship"*
  - Texas Christian University Annual Student Research Symposium | Fort Worth, TX | 2022
  - American Astronomical Society Winter Meeting | Seattle, WA | 2023
- *"Spectral Runway: An Analysis of Solar Balmer Lines through Observations and Models"*
  - American Geophysical Union Winter Meeting | New Orleans, LA | 2021

SKILLS

Computing: Python, JAX, Machine Learning, Data Analysis, Data Visualization, Software Development, High-Performance Computing, Git, Cloud Computing, GPUs